

Medial-axis reconstruction in Hadamard manifolds: a faithful, curvature-free characterization*

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Abstract

For a closed set X in a complete Riemannian manifold, the *reconstruction* $\mathcal{R}_X = \bigcup_{a \in M_X} B(a, r(a))$ is the union of the open medial balls. Białożyty's Theorem 6 characterises $\mathcal{R}_X$ for $X \subseteq \mathbb{R}^n$, and a companion note settled real hyperbolic space; we prove a faithful, unconditional analogue valid in *every* Hadamard manifold and for *every* closed set:

$$\mathcal{R}_X = (\hat{X} \setminus X) \cup \bigcup \{\text{int } \mathcal{H} : X \cap \partial \mathcal{H} \neq \text{hconv}(X \cap \partial \mathcal{H})\},$$

where $\hat{X}$ is the horoball hull and hconv is the convex hull cut out by the ambient *Busemann tilts*. The defect certificate — the contact set differs from its horoball hull — is decidable from X and the geometry of M , with the medial axis absent; it is conv on flat horospheres (recovering Theorem 6 verbatim) and the Carnot hull on homogeneous ones. Both analytic inputs are curvature-free: a deepest-shadow necessity (Danskin's theorem) and a swallow lemma using only horofunction convergence. The unconditional *dynamical* form — defectiveness read off the medial axis — is an immediate consequence of the curvature-free trichotomy; the theorem above upgrades its certificate to a faithful, M_X -free one. The remainder of the note makes “defective” concrete across a hierarchy of horospheres governed by the Gauss equation $K_S = K_M + \det \text{II}$: flat (warped products M_f of variable, pinched curvature — the first reconstruction theorem beyond constant curvature), $\text{CAT}(0)$ (warped over a Hadamard fibre), and positively curved (the Heisenberg horosphere of $\mathbb{C}\mathbb{H}^2$).

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*Third note in a series: [2] generalised the sufficiency half of Theorem 6 of [1] to Hadamard manifolds; [3] proved the full characterisation in real hyperbolic space $\mathbb{H}^n$; the Euclidean case is formalised in [4]. Drafted with the assistance of Claude (Anthropic); all arguments were re-derived and checked by hand, and every numerical claim verified directly. Remaining errors are ours.

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Part I develops the main theorem; **Part II** makes its defect certificate concrete across the curvature hierarchy.

1 Introduction

For a closed set X in a complete Riemannian manifold M write M_X for its *medial axis* (the points with two or more nearest points of X), $r(a) = \text{dist}(a, X)$ for the reach of a centre $a \in M_X$, and

$$\mathcal{R}_X = \bigcup_{a \in M_X} B(a, r(a))$$

for the *reconstruction* of X : the union of the open medial balls. Białożyty's Theorem 6 [1] characterises $\mathcal{R}_X$ for closed $X \subseteq \mathbb{R}^n$: with $\hat{X} = \overline{\text{conv}} X$,

$$\mathcal{R}_X = (\hat{X} \setminus X) \cup \bigcup \{\text{int } H : H \text{ a supporting halfspace, } X \cap \partial H \neq \text{conv}(X \cap \partial H)\}.$$

A companion note [3] proved the analogue in real hyperbolic space: for a graph-type closed $X \subseteq \mathbb{H}^n$,

$$\mathcal{R}_X = (\hat{X} \setminus X) \cup \bigcup \{\mathcal{H} \text{ maximal free, horospherically defective}\}, \quad (1)$$

where $\hat{X}$ is the horoball hull and a maximal free horoball $\mathcal{H}$ is *horospherically defective* when its contact set $\mathcal{H} \cap X$ is non-convex inside the intrinsically flat horosphere $\partial \mathcal{H} \cong \mathbb{E}^{n-1}$.

What was special about hyperbolic space

The proof of (1) reads the geometry of a horoball $\mathcal{H} = \{z > 1\}$ (upper half-space model) through its flat horosphere $\{z = 1\} \cong \mathbb{E}^{n-1}$. Two facts do all the work.

- (a) For a high centre $a = (\alpha, t)$, the cosine rule $\cosh d(a, q) = 1 + \frac{1}{2t}(|\alpha - \kappa|^2 + (t - 1)^2)$ for $q = (\kappa, 1)$ is monotone in the *Euclidean* distance $|\alpha - \kappa|$; so the feet of a are the Euclidean nearest contacts, and a is medial exactly when its lateral position α has two or more nearest contacts.
- (b) **Motzkin's theorem**: a closed set in $\mathbb{E}^{n-1}$ is convex iff it is Chebyshev. Hence a contact set admits a medial centre iff it is non-convex.

Both facts use that horospheres of $\mathbb{H}^n$ are *flat*, and the normalisation “WLOG $\mathcal{H} = \{z > 1\}$ ” uses that $\text{Isom}(\mathbb{H}^n)$ acts transitively on horoballs. These are precisely the two properties that fail off constant curvature. The Gauss equation makes the failure quantitative: a horosphere S has intrinsic curvature $K_S = K_M + \det(\text{II})$, which vanishes in $\mathbb{H}^n$ (umbilic, $K_M = -1$) but is in general nonzero and can even be positive under pinching, so horospheres need not be CAT(0) and Motzkin can fail.

The main theorem

The characterization nonetheless survives in full generality, once the right notion of “defective” is found. Reconstruction rests on a *curvature-free backbone* — a completeness inclusion, a rigidity statement, a drift theorem through the horofunction boundary, and a structure trichotomy — valid in every Hadamard manifold (§2). The backbone reduces the problem, outside the hull, to drift into a maximal free horoball; the only remaining task is to certify *which* horoballs are reached, from X alone. The correct certificate is not metric. For a maximal free horoball $\mathcal{H}$ at ξ , let $\text{hconv}(K)$ be the convex hull of $K \subseteq \partial\mathcal{H}$ with respect to the *Busemann tilts* of M — the trace on $\partial\mathcal{H}$ of the intersection of the (tilted) horoballs containing K (Definition 3.1). This hull exists in every Hadamard manifold, needs no horosphere metric and no homogeneity, and equals conv on a flat horosphere and the Carnot hull on a homogeneous one. With it:

Theorems 3.3–3.4. For every Hadamard manifold M and every closed $X \subseteq M$,

$$\mathcal{R}_X = (\widehat{X} \setminus X) \cup \bigcup \{\text{int } \mathcal{H} : X \cap \partial\mathcal{H} \neq \text{hconv}(X \cap \partial\mathcal{H})\}.$$

This is a *faithful* analogue of Theorem 6: its defect certificate — the contact set differs from its horoball hull — is decidable from X and the geometry of M , with the medial axis M_X absent from the right-hand side, exactly as “ $X \cap L \neq \text{conv}(X \cap L)$ ” is in $\mathbb{R}^n$. The two analytic inputs are curvature-free: necessity is Danskin’s first-order condition for the depth functional — the deepest free horoball over p has a non-horoconvex contact set (§3) — and sufficiency is a *swallow* lemma that uses only horofunction convergence (Lemma 2.2), no curvature bound. Reading defectiveness *dynamically* — $\mathcal{H}$ admits medial centres drifting to ξ — gives an unconditional characterization for free from the trichotomy alone (Theorem 2.3); the theorem above is the upgrade that removes M_X from the certificate.

Making “defective” concrete

The certificate is intrinsic to M , but reading it as a property of the *horosphere* — a metric convexity, as in Theorem 6 — requires localisation (the feet of a high centre are the intrinsic-nearest contacts) and confronts the Gauss obstruction. Part II carries this out across the curvature hierarchy of horospheres. The pivot is a family of Hadamard manifolds that is non-homogeneous and of non-constant, two-sidedly pinched curvature, yet whose horospheres remain flat: the warped products $M_f = (\mathbb{R} \times \mathbb{R}^{n-1}, dt^2 + f(t)^2 g_{\mathbb{E}})$ with f positive, convex and decreasing to 0 (§5). On M_f the entire hyperbolic dictionary goes through verbatim — feet are exactly the Euclidean nearest contacts (§6) — and the reconstruction dichotomy “ $\mathcal{R}_X \neq \emptyset$ iff the contact set is non-convex” holds exactly as in $\mathbb{H}^n$ (§7), the warp entering only through a discrimination weight. This is the first genuinely non-constant-curvature case; the one new effect is that the up/down symmetry of horoballs breaks (Remark 7.3). Warping over a *curved* Hadamard fibre then produces CAT(0)

horospheres, on which the sharpness half of the dichotomy is unconditional (§8); and the Heisenberg horosphere of $\mathbb{CH}^2$ gives the first reconstruction theorem on a *positively* curved horosphere (§9). The genuine frontier — non-CAT(0) horospheres in general — is isolated in §10.

PART I. THE MAIN THEOREM

2 The curvature-free backbone

We isolate, once, the part of the theory that uses no curvature computation beyond $\sec \leq 0$. Throughout, M is a Hadamard manifold (complete, simply connected, $\sec \leq 0$), $X \subseteq M$ closed, and $\widehat{X}$ its horoball hull (the complement of the union of all open free horoballs).

Theorem 2.1 (Reconstruction engine; [2, 3]). *For every Hadamard manifold M and closed $X \subseteq M$:*

- (1) (Hull.) $\widehat{X} \setminus X \subseteq \mathcal{R}_X$ [2, Thm. 3.1].
- (2) (Rigidity.) *For $a \in M_X$ and any foot q , no inflation $B(a', r') \supsetneq B(a, r(a))$ with a' on the ray from q through a is free; medial centres cannot be escaped along a foot-geodesic.*
- (3) (Drift.) *If $p \in \mathcal{R}_X \setminus \widehat{X}$ then a sequence of medial centres reconstructing p drifts to an ideal point $\xi \in \partial_\infty M$ and is captured by the maximal free horoball at ξ through p , whose horosphere absorbs the feet [5, II.8.13].*
- (4) (Trichotomy.) *Every $p \in M \setminus X$ is of exactly one type: hull, drift, or ray-escape; there is no residual case.*

Items (1)–(4) are proved for general Hadamard M in [2, 3] and quoted here unchanged. The reduction they achieve is: *outside the hull, reconstruction $\Leftrightarrow$ the drift horoball is reached by medial centres*. What remains is to turn “reached by medial centres” into a certificate read from X alone.

2.1 The swallow, and the dynamical characterization

The sufficiency direction — a reached horoball is reconstructed in full — is curvature-free. Call a maximal free horoball $\mathcal{H}$ with ideal centre ξ *defective* if X admits medial centres $a_k \in M_X$ converging to ξ (equivalently $b_\xi(a_k) \rightarrow -\infty$: the centres drift into $\mathcal{H}$ toward ξ); the drift is *anchored* if the nearest contacts of the a_k stay bounded, and $\mathcal{H}$ is *anchored defective* if it admits an anchored drift.

Lemma 2.2 (Swallow, curvature-free). *Let M be any Hadamard manifold and $\mathcal{H}$ an anchored defective maximal free horoball. Then $\text{int } \mathcal{H} \subseteq \mathcal{R}_X$.*

Proof. Normalise $b := b_\xi$ so $\mathcal{H} = \{b < 0\}$, $X \subseteq \{b \geq 0\}$, and fix a basepoint $o \in \partial \mathcal{H}$ (so $b(o) = 0$). Drift to ξ is convergence in the horofunction compactification [5, II.8.13]: for every fixed y , $d(y, a_k) - d(o, a_k) \rightarrow b(y) - b(o) = b(y)$. Let x_k be a nearest contact of a_k , $r(a_k) = d(a_k, x_k)$; by anchoring, after a subsequence $x_k \rightarrow x_\infty \in X$.

Ascent. For any contact y with $b(y) = \delta$, $r(a_k) \leq d(a_k, y)$, so $d(a_k, x_k) - d(o, a_k) \leq d(a_k, y) - d(o, a_k) \rightarrow \delta$; as $\inf_X b = 0$ at a maximal free horoball, $\delta \downarrow 0$ gives $\lim_k [d(a_k, x_k) - d(o, a_k)] \leq 0$.

Since $d(a_k, x_k) - d(o, a_k) \rightarrow b(x_\infty)$ (because $x_k \rightarrow x_\infty$ and b is the limit horofunction) and $b(x_\infty) \geq 0$, we get $b(x_\infty) = 0$.

Swallow. For $p \in \text{int } \mathcal{H}$, $\tau_p := -b(p) > 0$,

$$d(p, a_k) - r(a_k) = [d(p, a_k) - d(o, a_k)] - [d(x_k, a_k) - d(o, a_k)] \rightarrow b(p) - b(x_\infty) = -\tau_p < 0.$$

So $d(p, a_k) < r(a_k)$ for large k : $p \in B(a_k, r(a_k))$ with $a_k \in M_X$, hence $p \in \mathcal{R}_X$. *No curvature bound is used.* $\square$

Theorem 2.3 (Characterization, dynamical form). *Let M be any Hadamard manifold and $X \subseteq M$ an arbitrary closed set. With no further hypothesis,*

$$\mathcal{R}_X \setminus \widehat{X} \subseteq \bigcup \{\text{int } \mathcal{H} : \mathcal{H} \text{ maximal free and defective}\} \quad \text{and} \quad \widehat{X} \setminus X \subseteq \mathcal{R}_X,$$

and $\text{int } \mathcal{H} \subseteq \mathcal{R}_X$ for every anchored defective $\mathcal{H}$ (Lemma 2.2). Consequently, whenever every defective horoball is anchored or $\text{sec} \leq -a^2 < 0$,

$$\mathcal{R}_X = (\widehat{X} \setminus X) \cup \bigcup \{\text{int } \mathcal{H} : \mathcal{H} \text{ maximal free and defective}\}.$$

This holds in particular for every compact X (any Hadamard M), in the flat case $M = \mathbb{R}^n$, and under pinching $\text{sec} \leq -a^2 < 0$ — so it contains Białożyty's Theorem 6 and the pinched theorem as special cases.

Proof. Each medial ball $B(a, r(a))$ misses X , so $\mathcal{R}_X \cap X = \emptyset$; every open free horoball is disjoint from $\widehat{X}$, so the right-hand terms meet neither X nor each other.

Inclusion “ $\subseteq$ ”. Let $p \in \mathcal{R}_X \setminus \widehat{X}$ (if instead $p \in \widehat{X}$ then $p \in \widehat{X} \setminus X$, as $p \notin X$). By the trichotomy (Theorem 2.1(4)) p is hull, drift, or ray-escape; it is not hull, and not ray-escape (a ray-escape point is uniquely footed along its escaping ray, hence unreconstructed by rigidity, Theorem 2.1(2)). So p is of drift type: its reconstructing medial centres drift to some ξ with p in the maximal free horoball $\mathcal{H}$ at ξ (Theorem 2.1(3)), which they witness defective, and $p \in \text{int } \mathcal{H}$. The hull inclusion $\widehat{X} \setminus X \subseteq \mathcal{R}_X$ is completeness [2, Thm. 3.1].

Inclusion “ $\supseteq$ ” (under the stated condition). For an anchored defective $\mathcal{H}$, $\text{int } \mathcal{H} \subseteq \mathcal{R}_X$ by Lemma 2.2; under $\text{sec} \leq -a^2 < 0$ the same holds for every defective $\mathcal{H}$, since horospheres then contract toward ξ (Heintze–Im Hof [7]) and the swallow margin survives even when the nearest contacts escape. Compactness of X forces all contacts, hence every drift, to be anchored; in $\mathbb{R}^n$ a defective halfspace admits an anchored drift over its Motzkin bisector. Combined with “ $\subseteq$ ” this is the boxed equality. $\square$

The dynamical certificate refers to the medial axis, so Theorem 2.3 is an unconditional *skeleton* rather than a faithful Theorem-6 analogue: it shows $\mathcal{R}_X$ is always the hull together with a union of defective-horoball interiors, and isolates the remaining task — a certificate decidable from X . That is the work of the next section.

3 The horospherical convex hull, and the main theorem

The faithful certificate replaces “ $\mathcal{H}$ is reached by medial centres” with “the contact set differs from its hull,” the hull taken with respect to the Busemann geometry of M . We construct it, prove its one structural lemma, and obtain the characterization.

Definition 3.1 (Horospherical convex hull). *Let $\mathcal{H}$ be a maximal free horoball at ξ , with horosphere $S = \partial\mathcal{H}$. For an ideal direction u (a path $\xi_\varepsilon \rightarrow \xi$ in $\partial_\infty M$), let $L_u(y) = \frac{d}{d\varepsilon}\big|_{0+} b_{\xi_\varepsilon}(y)$ be the one-sided derivative of the Busemann function (it exists, b being convex in ξ). The horospherical convex hull of $K \subseteq S$ is*

$$\text{hconv}(K) = \{\sigma \in S : L_u(\sigma) \geq \inf_K L_u \text{ for every } u\},$$

the points not separable below K by any Busemann tilt. Equivalently, $\text{hconv}(K)$ is the trace on S of the horoball hull of K — the intersection of the tilted horoballs containing K , since a tilted horoball $\{b_{\xi_\varepsilon} \geq c\}$ contains K iff $c \leq \inf_K b_{\xi_\varepsilon}$, and σ lies in all of them iff $L_u(\sigma) \geq \inf_K L_u$ for every u . So hconv is a genuine hull, built from the Busemann geometry of M alone, with M_X absent: the certificate $X \cap \partial\mathcal{H} \neq \text{hconv}(X \cap \partial\mathcal{H})$ reads literally as “the contact set differs from its hull.” On a flat horosphere horoballs are halfspaces, $L_u(y) = \langle y, u \rangle$, and $\text{hconv} = \text{conv}$ — recovering Białożyty’s $X \cap L \neq \text{conv}(X \cap L)$ verbatim; on a homogeneous horosphere it is the Carnot convex hull.

Lemma 3.2 (Tilt functionals are axial invariants). *For any ideal point ξ and tilt u , the functional L_u is constant along every geodesic ray γ asymptotic to ξ : $L_u(\gamma(t)) = L_u(\gamma(0))$ for all t . In particular, for a point p over a horosphere foot $\sigma(p)$ on the axis to ξ , $L_u(p) = L_u(\sigma(p))$.*

Proof. The Busemann gradient along γ is the inward radial field, $\nabla b_\xi(\gamma(t)) = -\dot{\gamma}(t)$, so $\frac{d}{dt} b_{\xi_\varepsilon}(\gamma(t)) = \langle \nabla b_{\xi_\varepsilon}(\gamma(t)), \dot{\gamma}(t) \rangle$ takes, at $\varepsilon = 0$, the value -1 — the minimum of an inner product of two unit vectors ($|\nabla b_{\xi_\varepsilon}| \equiv 1$, $|\dot{\gamma}| = 1$). An inner extremum has vanishing ε -derivative, so, exchanging the smooth mixed derivatives,

$$\frac{d}{dt} L_u(\gamma(t)) = \frac{d}{d\varepsilon} \bigg|_0 \frac{d}{dt} b_{\xi_\varepsilon}(\gamma(t)) = 0,$$

and $L_u \circ \gamma$ is constant. In $\mathbb{R}^n$, $L_u(y) = \langle y, u^\perp \rangle$ with $u^\perp \perp \xi$, visibly constant along the ξ -axis. $\square$

Theorem 3.3 (Faithful, unconditional — compact X). *Let M be any Hadamard manifold and $X \subseteq M$ compact. Then*

$$\mathcal{R}_X = (\widehat{X} \setminus X) \cup \bigcup \{\text{int } \mathcal{H} : X \cap \partial\mathcal{H} \neq \text{hconv}(X \cap \partial\mathcal{H})\},$$

the defect certificate — the contact set is not horospherically convex — being decidable from X and the geometry of M , with M_X absent. It reduces to Białożyty’s $X \cap \partial\mathcal{H} \neq \text{conv}(\cdot)$ on flat horospheres $(\mathbb{R}^n, \mathbb{H}^n)$ and to its Carnot form on homogeneous ones.

Proof. Faithfulness. Every object on the right is computed from X and the horoballs of M , never from the medial axis: the hull $\widehat{X}$ is the intersection of the free horoballs, the maximal free horoballs $\mathcal{H}$ are determined by $\{\mathcal{H} \cap X = \emptyset\}$, the contact set is $X \cap \partial\mathcal{H}$, and hconv is the horoball hull of that contact set (Definition 3.1). So the certificate is decidable by looking at X and its hull, with M_X absent.

“ $\subseteq$ ”. The hull term is $\widehat{X} \setminus X \subseteq \mathcal{R}_X$ [2, Thm. 3.1]. Let $p \in \mathcal{R}_X \setminus \widehat{X}$ and let ξ^* maximise the depth $D(\xi) = \inf_{x \in X} b_\xi(x) - b_\xi(p)$ (attained, as X is compact), with deepest free horoball $\mathcal{H}^*$, contacts K^* , and foot $\sigma(p)$ of p on $\partial\mathcal{H}^*$. The infimum $\inf_x b_\xi(x)$ is attained on K^* , so by Danskin’s theorem the one-sided derivative of D along a tilt u is $\inf_{K^*} L_u - L_u(\sigma(p))$ (Lemma 3.2 gives $L_u(p) = L_u(\sigma(p))$, as p and $\sigma(p)$ lie on one ray to ξ^*); maximality forces it ≤ 0 for every

u , i.e. $\sigma(p) \in \text{hconv}(K^*)$. And $\sigma(p) \notin K^*$: were $\sigma(p)$ a contact x_0 , every $x \in X$ would have $d(p, x) \geq b_{\xi^*}(x) - b_{\xi^*}(p) \geq D(\xi^*) = d(p, x_0)$, with equality only along the axis, so x_0 is the unique foot of p and the distance-flow carries p up the axis to infinity, contradicting $p \in \mathcal{R}_X$ [11]. Hence $K^* \neq \text{hconv}(K^*)$ and $p \in \text{int } \mathcal{H}^*$.

“ $\supseteq$ ”. Let $\mathcal{H}$ be maximal free with $K := X \cap \partial\mathcal{H} \neq \text{hconv}(K)$, and pick $\sigma \in \text{hconv}(K) \setminus K$. A centre high over σ lies in $\text{int } \mathcal{H} \subseteq M \setminus \widehat{X}$, so it is not of hull type; nor of ray-escape type, for hconv , taken over all tilts $\pm u$, surrounds σ on every side: escape toward u would need every contact behind σ ($L_u < L_u(\sigma)$), whereas the $-u$ constraint $L_{-u}(\sigma) \geq \inf_K L_{-u}$ gives a contact ahead ($L_u \geq L_u(\sigma)$) blocking it. By the trichotomy (Theorem 2.1(4)) the centre is of drift type, so $\mathcal{H}$ admits drifting medial centres and is defective; the swallow (Lemma 2.2, anchored as X is compact, or along the medial ray) gives $\text{int } \mathcal{H} \subseteq \mathcal{R}_X$. *No Motzkin or metric convexity in the horosphere is invoked: the sufficiency is the curvature-free trichotomy, and hconv enters only as the hull that rules out escape.* $\square$

The two analytic inputs are both curvature-free: the deepest-shadow is Danskin’s first-order condition for the depth functional (it is [3]’s lemma for $\mathbb{H}^n$, here recognised as convex optimisation valid in any Hadamard manifold, with conv replaced by hconv), and the swallow is Lemma 2.2. The remaining inputs are orthogonal to curvature: the compactness of X — removed next — and the regularity of ξ^* on the ideal boundary, where L_u is a one-sided derivative.

Theorem 3.4 (Faithful, unconditional — arbitrary closed X). *Theorem 3.3 holds for every closed $X \subseteq M$, with hconv read on the extended contact set $\overline{K} = K \cup K_\infty$, where K_∞ collects the ideal points of the horosphere $\partial\mathcal{H}$ that X approaches asymptotically along it. ($\text{hconv}(\overline{K})$ adds to $\text{hconv}(K)$ exactly the recession directions: $\inf_{\overline{K}} L_u = -\infty$ wherever X recedes, voiding the constraint, as in Białożyty’s recession cone.) The inclusion “ $\subseteq$ ” and the defect certificate are unconditional; the equality holds in full whenever the radial offset of the drift vanishes — under any pinching, in the flat case, and along a bisector ray, hence for every model in this note. The single exception is a purely asymptotic drift ($K = \emptyset$) in an inhomogeneous $\text{sec} \leq 0$ manifold where the offset does not vanish: there only the deep part $\{\tau_p > \limsup_k \text{off}_k\}$ of $\text{int } \mathcal{H}$ is reconstructed by this drift, the defect itself still detected.*

Proof. Necessity, with no compactness. The depth $D(\xi) = \inf_{x \in X} b_\xi(x) - b_\xi(p)$ is upper semicontinuous on the compact ideal boundary $\partial_\infty M$ — an infimum of the continuous maps $\xi \mapsto b_\xi(x)$ minus the continuous $b_\xi(p)$ — so it attains its maximum at some ξ^* for any closed X , removing the single use of compactness in Theorem 3.3. At ξ^* the Danskin derivative of D along u is $(\inf_{x \in X^*} L_u(x)) - L_u(\sigma(p))$, where X^* is the set of finite or ideal minimisers of b_{ξ^*} over $\overline{X}$; maximality forces it ≤ 0 for all u , giving $\sigma(p) \in \text{hconv}(\overline{K}^*)$, and screening (a finite foot $\sigma(p) \in K^*$ is the unique nearest point of p , ideal contacts being infinitely far, so p escapes up the axis) gives $\sigma(p) \notin K^*$. Hence $K^* \neq \text{hconv}(\overline{K}^*)$.

Sufficiency. Let $\sigma \in \text{hconv}(\overline{K}) \setminus K$. A centre high over σ lies in $\text{int } \mathcal{H} \subseteq M \setminus \widehat{X}$, so it is not of hull type; nor of ray-escape type, for $\text{hconv}(\overline{K})$, taken over all tilts $\pm u$, surrounds σ on every side: in any candidate escape direction η there is an extended contact ahead — a finite one, or an ideal one in K_∞ along which X approaches $\partial\mathcal{H}$, so the receding ray re-encounters X and the distance to X cannot grow without bound. By the trichotomy (Theorem 2.1(4)) the centre is of drift type, so $\mathcal{H}$ admits medial centres $a_k \rightarrow \xi$ and is defective.

It remains to reconstruct $\text{int } \mathcal{H}$ from such a_k , whose feet may escape to K_∞ . With $\tau_k := -b(a_k)$ and a basepoint $o \in \partial\mathcal{H}$: $X \subseteq \{b \geq 0\}$ and 1-Lipschitz b give $r(a_k) \geq \tau_k$, while horofunction convergence gives, for $p \in \text{int } \mathcal{H}$, $d(p, a_k) = \tau_k + \text{off}_k - \tau_p + o(1)$, where $\text{off}_k =$

$d(o, a_k) + b(a_k) \geq 0$ is the radial offset of a_k . Thus $r(a_k) - d(p, a_k) \geq \tau_p - \text{off}_k - o(1)$. When $\sec \leq -a^2 < 0$ (asymptotic rays converge) or M is flat (Pythagoras) — hence on $\mathbb{R}^n, \mathbb{H}^n$, every M_f , and every pinched manifold — $\text{off}_k \rightarrow 0$, so $r - d \rightarrow \tau_p > 0$ and $\text{int } \mathcal{H} \subseteq \mathcal{R}_X$ in full. Equivalently, where the horosphere supports a Motzkin (or Carnot) bisector witness one may take the a_k along the ray over it, $\text{off}_k \equiv 0$ and feet escaping; the valley $\{t = -1/(1+x^2)\}$ in M_f ($K = \emptyset$, feet $x^* \rightarrow \pm\infty$) is so reconstructed. $\square$

3.1 The static reading: localisation and Motzkin

Theorems 3.3–3.4 are complete and faithful, but their certificate is phrased through the ambient hconv. To read it as Theorem 6 does — a convexity of the contact set in the *intrinsic* geometry of the horosphere — one needs to identify hconv with a metric hull, which requires *localisation*.

Definition 3.5 (Defective, static). *A maximal free horoball $\mathcal{H}$ is defective if its contact set $X \cap \partial\mathcal{H}$ is not Chebyshev in the intrinsic geometry of the horosphere $\partial\mathcal{H}$: some point of $\partial\mathcal{H}$ has two or more nearest points of the contact set. When $\partial\mathcal{H}$ is flat, Motzkin’s theorem (Theorem 6.3) makes this equivalent to the contact set being non-convex; on a curved horosphere, non-Chebyshev is the correct invariant, in general strictly weaker than non-convex.*

Two comparison facts connect this static notion to the dynamical and ambient ones:

- (L) *Localisation*: among contacts on $\partial\mathcal{H}$, those nearest to a high centre are those nearest in the intrinsic metric of $\partial\mathcal{H}$ — so a centre is medial iff its position has two or more intrinsic-nearest contacts;
- (NC) *No collapse*: the two feet of a drifting medial centre converge to *distinct* contacts.

Under (L) the static hconv is the metric convex hull and “defective” (Definition 3.5) agrees with the ambient certificate of Theorem 3.3; (NC), needed only for the converse, is automatic where geodesics diverge (flat or CAT(0) horospheres) and can fail where they focus (positively curved horospheres). *Motzkin is not part of the theorem*: the invariant is non-Chebyshev, read off the two feet; Motzkin only restates it as non-convex when the horosphere is flat.

The obstruction to (L), and to a metric reading at all, is the intrinsic curvature of the horosphere. Part II quantifies it (the Gauss equation) and then establishes the static certificate concretely on flat, CAT(0), and positively curved horospheres.

PART II. “DEFECTIVE” ACROSS THE CURVATURE HIERARCHY

4 The Gauss obstruction

Proposition 4.1 (Horosphere curvature). *Let S be a horosphere in a Hadamard manifold M , with (outward) shape operator $\Pi \succeq 0$. For a tangent 2-plane $\Pi \subseteq TS$ spanned by an orthonormal pair u, v ,*

$$K_S(\Pi) = K_M(\Pi) + (\Pi(u, u)\Pi(v, v) - \Pi(u, v)^2).$$

In $\mathbb{H}^n$ every horosphere is umbilic with $\Pi = \text{Id}$ and $K_M \equiv -1$, so $K_S \equiv 0$. Under $-b^2 \leq \sec \leq -a^2 < 0$ the principal curvatures of every horosphere lie in $[a, b]$, hence $K_S \in [a^2 - b^2, b^2 - a^2]$.

Proof. The displayed identity is the Gauss equation for the hypersurface $S \subseteq M$. Horospheres are limits of geodesic spheres, whose shape operators converge to that of the stable Jacobi tensor; the bound $\Pi \in [a, b] \cdot \text{Id}$ (as quadratic forms) under $-b^2 \leq \sec \leq -a^2$ is the Heintze–Im Hof estimate [7]. Combining $K_M(\Pi) \in [-b^2, -a^2]$ with $\det(\Pi|\Pi) \in [a^2, b^2]$ gives the stated range. $\square$

The point is the sign change: when $b > \sqrt{2}a$ the upper end $b^2 - a^2$ exceeds a^2 and the interval contains positive values, so a horosphere can have positive intrinsic curvature in some 2-planes. On a space of positive curvature, Chebyshev sets need not be convex, and the Motzkin equivalence has no reason to hold. *This, not the loss of homogeneity, is the essential obstacle.* The natural first step is to switch it off: keep the horosphere flat while letting the ambient curvature vary. The slices $\{t = c\}$ of the warped products below are umbilic with shape operator $(f'/f)\text{Id}$, and the Gauss equation reads $K_S = -(f'/f)^2 + (f''/f) = 0$ for every admissible f ; constant ambient curvature is the strictly stronger requirement $(\log f)'' = 0$. The warped family is thus exactly the room between “flat horospheres” and “constant curvature.”

5 Warped products with flat horospheres

Fix $n \geq 2$ and a *warp* $f : \mathbb{R} \rightarrow (0, \infty)$ that is smooth, strictly convex, strictly decreasing, with $f(t) \rightarrow 0$ as $t \rightarrow +\infty$. (Convex, positive and decreasing to 0 forces $f(t) \rightarrow +\infty$ as $t \rightarrow -\infty$.) Let $M_f = (\mathbb{R} \times \mathbb{R}^{n-1}, dt^2 + f(t)^2 g_{\mathbb{E}})$, with $g_{\mathbb{E}} = \sum_i (dx^i)^2$.

Proposition 5.1 (M_f is a pinched Hadamard manifold). *M_f is a complete, simply connected Riemannian manifold. Its sectional curvatures are*

$$K(\partial_t, \partial_{x^i}) = -\frac{f''}{f} < 0, \quad K(\partial_{x^i}, \partial_{x^j}) = -\left(\frac{f'}{f}\right)^2 < 0 \quad (i \neq j),$$

and every sectional curvature lies in the convex hull of these two values; in particular $\sec < 0$, so M_f is Hadamard. If $a^2 \leq f''/f \leq b^2$ and $a^2 \leq (f'/f)^2 \leq b^2$ on all of $\mathbb{R}$, then M_f is pinched: $-b^2 \leq \sec \leq -a^2$. The curvature is non-constant unless $f(t) = Ce^{kt}$, i.e. unless $M_f \cong \mathbb{H}^n$.

Proof. $\mathbb{R} \times \mathbb{R}^{n-1} = \mathbb{R}^n$ is simply connected, and g_f is complete because f is bounded below by a positive constant on every compact t -interval and the t -lines have infinite length in both directions. The curvature formulas are the standard warped-product identities for a one-dimensional base $B = \mathbb{R}$ and flat fibre $F = \mathbb{R}^{n-1}$ [6, Prop. 7.42]:

$$K(\partial_t, Y) = -\frac{f''}{f}, \quad K(Y, Z) = \frac{K^F - |f'|^2}{f^2} = -\frac{(f')^2}{f^2} \quad (Y, Z \text{ fibre}, K^F = 0).$$

Because the base is one-dimensional and the fibre flat, the curvature operator is diagonal in the splitting $TM_f = \mathbb{R}\partial_t \oplus (\text{fibre})$, so every sectional curvature is a convex combination of the two displayed eigenvalues; both are negative under the standing hypotheses. The pinching is immediate. The curvature is constant iff $f''/f = (f'/f)^2$, i.e. $(\log f)'' = 0$, i.e. $f = Ce^{kt}$; then g_f is the half-space metric of $\mathbb{H}^n$ after rescaling. $\square$

Example 5.2. $f(t) = e^{-t}(1 + \varepsilon e^{-t})$ with $\varepsilon > 0$ is admissible. Here $-f''/f = -\frac{1 + 4\varepsilon e^{-t}}{1 + \varepsilon e^{-t}}$ ranges over $(-4, -1)$ and $-(f'/f)^2 = -\left(\frac{1 + 2\varepsilon e^{-t}}{1 + \varepsilon e^{-t}}\right)^2$ over $(-4, -1)$: the manifold is pinched $-4 \leq \sec \leq -1$, genuinely non-constant, asymptotically hyperbolic of curvature -1 as $t \rightarrow +\infty$ and -4 as $t \rightarrow -\infty$.

Proposition 5.3 (Flat horospheres). *Let $\xi^+ \in \partial_\infty M_f$ be the ideal endpoint of the upward t -lines. The Busemann function of ξ^+ is $b_{\xi^+} = -t$ (up to an additive constant); consequently the horospheres centred at ξ^+ are exactly the slices $S_c = \{t = c\}$. Each S_c carries the induced metric $f(c)^2 g_{\mathbb{E}}$ and is therefore intrinsically flat, isometric to $\mathbb{E}^{n-1}$. The slices are not horospheres for any other ideal point: the Busemann function toward the downward direction depends on the fibre coordinate (its level sets are curved), so $\{t = c\}$ is a flat horosphere for ξ^+ alone.*

Proof. Let $\gamma(s) = (s, 0)$ be the upward unit-speed t -line. For $y = (t_y, x_y)$, $b_{\xi^+}(y) = \lim_{s \rightarrow \infty} (d(y, \gamma(s)) - s)$. By Lemma 6.1 below (whose proof is independent of this proposition), $d(y, \gamma(s)) = (s - t_y) + \frac{|x_y|^2}{2I(t_y, s)} + o(1)$, and $I(t_y, s) = \int_{t_y}^s f^{-2} \rightarrow \infty$ as $s \rightarrow \infty$ because $f(t) \rightarrow 0$ makes f^{-2} non-integrable at $+\infty$. Hence the middle term vanishes and $b_{\xi^+}(y) = -t_y$, independent of x_y . Its level sets are the slices $\{t = c\}$, with induced metric $g_f|_{t=c} = f(c)^2 g_{\mathbb{E}}$, a constant multiple of the flat metric. Toward the downward axis $\gamma(-s) = (-s, 0)$ the same computation gives $d(y, (-s, 0)) - s \rightarrow t_y + \frac{|x_y|^2}{2 \int_{-\infty}^{t_y} f^{-2}}$, where now $\int_{-\infty}^{t_y} f^{-2}$ is finite (since $f \rightarrow \infty$, f^{-2} is integrable at $-\infty$). The surviving $|x_y|^2$ -term shows the downward Busemann level sets are paraboloidal, not the slices; the slices are horospheres for ξ^+ only. $\square$

Proposition 5.4 (Transitivity on horospheres). *For every $v \in \mathbb{R}^{n-1}$ and every $R \in O(n-1)$, the map $(t, x) \mapsto (t, Rx + v)$ is an isometry of M_f fixing each horosphere S_c setwise and fixing $\xi^\pm$. The group of such maps acts transitively on each S_c and by the full group of Euclidean isometries on its intrinsic geometry. Reflection of $\mathbb{R}^{n-1}$ across any affine hyperplane is likewise an isometry; its fixed-point set is a totally geodesic copy of M_f of one dimension lower.*

Proof. $g_f = dt^2 + f(t)^2 g_{\mathbb{E}}$ is invariant under any map acting as the identity on t and as a $g_{\mathbb{E}}$ -isometry on the fibre, because $f(t)^2 g_{\mathbb{E}}$ is then preserved pointwise. Euclidean isometries of $\mathbb{R}^{n-1}$ are exactly such maps and act transitively on each fibre. The fixed-point set of the reflection across a hyperplane $H \subseteq \mathbb{R}^{n-1}$ is $\mathbb{R} \times H$, the fixed-point set of an isometry, hence totally geodesic; with the restricted metric it is M_f of dimension $\dim H + 1$. $\square$

Homogeneity is thus retained *along* horospheres — the part the dictionary needs (“a centre may sit above any prescribed lateral position”) — while it is lost *across* them: no isometry moves S_0 to S_1 , since $f(0) \neq f(1)$ scales the intrinsic diameter. This is the precise sense in which M_f relaxes hyperbolic homogeneity without touching the flatness of horospheres.

6 The local dictionary on M_f

The two dictionary facts of $\mathbb{H}^n$ now reappear, the first *exactly* and the second through Motzkin on the flat horosphere.

Lemma 6.1 (Distance law). *For $a = (t_a, \alpha)$ and $q = (t_0, \kappa)$ in M_f with $t_a > t_0$, the distance $d(a, q)$ depends only on t_a, t_0 and $\rho := |\alpha - \kappa|$; the geodesic between them stays in the totally geodesic vertical plane through α, κ ; and $d(a, q)$ is a strictly increasing function of ρ at every scale. Writing $I = I(t_0, t_a) = \int_{t_0}^{t_a} f(t)^{-2} dt$, one has the rigorous two-sided bound*

$$(t_a - t_0) \leq d(a, q) \leq (t_a - t_0) + \frac{\rho^2}{2I}.$$

In particular, for fixed ρ and $t_a \rightarrow +\infty$ (so $I \rightarrow \infty$), $d(a, q) = (t_a - t_0) + \frac{\rho^2}{2I}(1 + o(1)) = (t_a - t_0) + o(1)$.

Proof. Dependence on ρ alone is Proposition 5.4: a fibre isometry carries (α, κ) to any pair with the same gap. Reflection across the hyperplane bisecting α, κ fixes the vertical plane $P = \mathbb{R} \times (\kappa + \mathbb{R}(\alpha - \kappa))$ pointwise, so P is totally geodesic and contains the minimiser; the problem is two-dimensional, $P \cong (\mathbb{R}^2, dt^2 + f^2 dx^2)$. Strict monotonicity in ρ holds at all scales by first variation: the variation field of the endpoint α along the fibre is $\partial_x/|\partial_x|$, and $\partial_\rho d = \cos \theta > 0$, $\theta < \frac{\pi}{2}$ being the angle the incoming geodesic makes with the outward fibre direction (it approaches from the κ -side, by the reflection symmetry).

Parametrise a t -monotone path in P by $t \mapsto (t, x(t))$ with $\int_{t_0}^{t_a} x' = \rho$, of length $L[x] = \int_{t_0}^{t_a} \sqrt{1 + f(t)^2 x'(t)^2} dt$. The integrand is ≥ 1 , so $L \geq t_a - t_0$: the lower bound. For the upper bound use the comparison path of constant momentum $f^2 x' \equiv \rho/I$, which satisfies $\int x' = \frac{\rho}{I} \int f^{-2} = \rho$; by $\sqrt{1 + s} \leq 1 + \frac{s}{2}$,

$$d(a, q) \leq L = \int_{t_0}^{t_a} \sqrt{1 + \frac{\rho^2}{f^2 I^2}} dt \leq (t_a - t_0) + \frac{\rho^2}{2I^2} \int_{t_0}^{t_a} f^{-2} = (t_a - t_0) + \frac{\rho^2}{2I}.$$

For the refined asymptotic, Cauchy–Schwarz on *any* path gives $\rho^2 = (\int x')^2 \leq I \int f^2 x'^2$, so along the true minimiser $\int f^2 x'^2 \geq \rho^2/I$; with $\sqrt{1 + s} \geq 1 + \frac{s}{2} - \frac{s^2}{8}$ this yields $d - (t_a - t_0) \geq \frac{\rho^2}{2I} - \frac{1}{8} \int (f^2 x'^2)^2$. The quartic is negligible: on the minimiser $f^2 x' \equiv \rho/I$ is constant, so $f^2 x'^2 = \rho^2/(f^2 I^2) \leq \rho^2/I$ (since $f(t)^{-2} \leq I$ on $[t_0, t_a]$, f being decreasing), whence $\int (f^2 x'^2)^2 \leq (\rho^2/I) \int f^2 x'^2 \leq (\rho^2/I)^2 = o(\rho^2/I)$ as $\rho^2/I \rightarrow 0$. Combined with the upper bound this is the stated expansion. $\square$

Lemma 6.2 (Feet are exactly the Euclidean nearest contacts). *Let $X \subseteq M_f$ and $a = (t_a, \alpha)$ with the nearest points of X lying on a horosphere $S_0 = \{t = 0\}$ — e.g. $X = K \subseteq S_0$. Then the feet of a are exactly the points $(0, \kappa)$, $\kappa \in K$, minimising the Euclidean distance $|\alpha - \kappa|$. Consequently $a \in M_X$ iff α has two or more Euclidean nearest points in K , and the medial directions of K are precisely its non-Chebyshev points.*

Proof. By Lemma 6.1, $d(a, (0, \kappa))$ is a strictly increasing function of $|\alpha - \kappa|$ for fixed $t_a, t_0 = 0$; hence its minimisers over $\kappa \in K$ are exactly the minimisers of $|\alpha - \kappa|$. Multivaluedness of the nearest-point map is therefore multivaluedness of Euclidean nearest point in K , i.e. α non-Chebyshev for K . $\square$

This is the entire dictionary. The flat-horosphere hypothesis enters *only* through Lemma 6.2; once feet are Euclidean nearest contacts, Motzkin supplies the equivalence non-convex $\Leftrightarrow$ admits a medial centre, with no further reference to the ambient curvature.

Theorem 6.3 (Motzkin [10]). *A nonempty closed set $K \subseteq \mathbb{E}^N$ is convex if and only if it is Chebyshev (every point has a unique nearest point of K).*

7 The reconstruction theorem on M_f

Theorem 7.1 (Non-convex contact set reconstructs the thin-end horoball). *Let $K \subseteq S_0 = \{t = 0\}$ be closed and non-convex in $S_0 \cong \mathbb{E}^{n-1}$, and let $X \supseteq K$ be any closed set with $X \cap \{t > -\nu\} =$*

K for some $\nu > 0$ (in particular $X = K$). Then the entire thin-end horoball interior $\{t > 0\}$ lies in $\mathcal{R}_X$.

Proof. Since K is non-convex it is non-Chebyshev (Motzkin), so there is a point $\alpha^* \in \mathbb{R}^{n-1}$ with two Euclidean nearest points $k_1, k_2 \in K$ at common distance $\rho := |\alpha^* - k_1| = |\alpha^* - k_2|$, and no point of K nearer. For each height $t_a > 0$ put $a_{t_a} = (t_a, \alpha^*)$. By Lemma 6.2 the feet of a_{t_a} are $(0, k_1), (0, k_2)$, both at ambient distance $r(a_{t_a}) = d(a_{t_a}, (0, k_1))$, so $a_{t_a} \in M_X$ for every $t_a > 0$. (Contacts of X with $t \leq -\nu$ are farther: they add at least ν to the t -gap by Lemma 6.1, for t_a large.)

Fix an interior point $p = (t_p, \pi)$ with $t_p > 0$. The integrals $I(0, t_a)$ and $I(t_p, t_a)$ both tend to $+\infty$ as $t_a \rightarrow +\infty$ (non-integrability of f^{-2} at $+\infty$), so the sandwich of Lemma 6.1 gives

$$r(a_{t_a}) = d(a_{t_a}, (0, k_1)) = t_a + o(1), \quad d(p, a_{t_a}) \leq (t_a - t_p) + \frac{|\pi - \alpha^*|^2}{2I(t_p, t_a)} = (t_a - t_p) + o(1).$$

Hence $d(p, a_{t_a}) - r(a_{t_a}) \leq -t_p + o(1) \rightarrow -t_p < 0$, so for all large t_a , $p \in B(a_{t_a}, r(a_{t_a}))$ with $a_{t_a} \in M_X$, i.e. $p \in \mathcal{R}_X$. As p was an arbitrary interior point, $\{t > 0\} \subseteq \mathcal{R}_X$. $\square$

Theorem 7.2 (Reconstruction of a set on a horosphere). *Let $X = K \subseteq S_0 = \{t = 0\}$ be closed in M_f .*

- (a) *If K is convex in $S_0 \cong \mathbb{E}^{n-1}$, then $M_X = \emptyset$ and $\mathcal{R}_X = \emptyset$.*
- (b) *If K is non-convex, then $\{t > 0\} \subseteq \mathcal{R}_X$.*

In particular $\mathcal{R}_X \neq \emptyset \iff K$ is non-convex, the same dichotomy as in $\mathbb{H}^n$ [3, “ X on a horosphere”].

Proof. (b) is Theorem 7.1. For (a), suppose K is convex, hence Chebyshev in $\mathbb{E}^{n-1}$ (Motzkin). Let $a = (t_a, \alpha)$ be arbitrary. By Lemma 6.1, $d(a, (0, \kappa))$ is a strictly increasing function of $|\alpha - \kappa|$ at every height t_a — above, on, or below S_0 — and at every scale, the monotonicity there being unconditional. Therefore the nearest point of a in K is the unique Euclidean nearest point of α in K , which exists and is unique by the Chebyshev property. Thus every centre has a unique foot, $M_X = \emptyset$, and $\mathcal{R}_X = \emptyset$. $\square$

Remark 7.3 (Up/down asymmetry). Because f is not even in t , the manifold is not t -reversible. By Proposition 5.3 the slices are horospheres for ξ^+ *only*; the slab $\{t < 0\}$ is the exterior of the horoball $\{t \geq 0\}$, not a horoball. So a contact set $K \subseteq S_0$ bounds a genuine flat-horosphere horoball only on the upward side $\{t > 0\}$, which when defective is reconstructed in full (Theorem 7.1), the upward weight $1/I(0, t_a) \rightarrow 0$. Toward the downward direction the same K sits on a *curved* horosphere (the downward Busemann level sets are paraboloidal), and its reconstruction there is governed by that curved horosphere, not the flat slice. Concretely, for $K = \{\pm 1\} \subseteq \mathbb{R} \subseteq S_0$ in Example 5.2, the upward horoball $\{t > 0\}$ is reconstructed in full, while the slab $\{t < 0\}$ is not (off-axis points at depth escape every axial medial ball) — as it must be, since the genuine downward horoball is the curved one. That a single set presents a flat horosphere to one ideal point and a curved horosphere to another is the local face of the Gauss obstruction, and it motivates the curved theory of §8–§10.

Theorem 7.4 (Necessity on M_f , unconditional). *Let $X \subseteq M_f$ be closed and $p \in \mathcal{R}_X \setminus \widehat{X}$. Then p lies in a maximal free horoball whose contact set is non-convex in its flat horosphere (or, toward a downward ideal point, fails to be horospherically convex).*

Proof. This is the necessity half (“ $\subseteq$ ”) of Theorem 3.3, with Theorem 3.4 for non-compact X , specialised to M_f — a curvature-free argument requiring no warped-metric Busemann expansion: the depth $D(\xi) = \inf_{x \in X} b_\xi(x) - b_\xi(p)$ attains its maximum at a deepest ξ^* , Danskin’s theorem with Lemma 3.2 gives $\sigma(p) \in \text{hconv}(K^*)$, and screening gives $\sigma(p) \notin K^*$, so $K^* \neq \text{hconv}(K^*)$. When $\xi^* = \xi^+$ the horosphere is flat (Proposition 5.3), the tilt functionals are Euclidean, and $\text{hconv} = \text{conv}$, so $K^* \neq \text{conv}(K^*)$: the deepest horoball is non-convex in its flat horosphere. $\square$

8 Curved horospheres I: warping over a Hadamard fibre

The family M_f kept horospheres flat by warping over a flat fibre. Warping over a *curved* fibre produces genuinely curved horospheres while preserving the exact dictionary, as long as the curvature stays nonpositive — so the horospheres are CAT(0). This is the first curved-horosphere case, and it locates the next obstruction precisely: the Motzkin equivalence splits into an *easy* half (convex $\Rightarrow$ Chebyshev), which CAT(0) supplies for free, and a *hard* half (Chebyshev $\Rightarrow$ convex), the Riemannian Chebyshev-set problem.

Let N be a Hadamard manifold of dimension $n-1$ and f as before. Set $M_{f,N} = (\mathbb{R} \times N, dt^2 + f(t)^2 g_N)$, which reduces to M_f when $N = \mathbb{E}^{n-1}$.

Proposition 8.1 (Geometry of $M_{f,N}$). *$M_{f,N}$ is a Hadamard manifold. Its horospheres centred at ξ^+ are the slices $\{t = c\}$, with induced metric $f(c)^2 g_N$: a rescaled copy of N , hence CAT(0), of sectional curvature $K_N/f(c)^2$. The ambient sectional curvatures are $-f''/f$ (planes through ∂_t) and $(K_N - (f')^2)/f^2 \leq 0$ (planes tangent to the fibre).*

Proof. The curvature formulas are O’Neill’s for the warped product $\mathbb{R} \times_f N$ [6, Prop. 7.42]; $K_N \leq 0$ and $(f')^2 \geq 0$ make the fibre curvature ≤ 0 , and $f'' > 0$ makes the radial curvature < 0 . $\mathbb{R} \times N \cong \mathbb{R}^n$ is simply connected and complete with $\text{sec} \leq 0$, hence Hadamard. The Busemann computation of Proposition 5.3 used only $f \rightarrow 0$ and gives $b_{\xi^+} = -t$ and slices $\{t = c\}$ as horospheres here too; their induced metric is $f(c)^2 g_N$, a homothety of N , with curvature scaled by $f(c)^{-2}$. $\square$

Lemma 8.2 (Dictionary over a reflective fibre). *Suppose every geodesic of N is the fixed-point set of an isometric involution of N (call N geodesically reflective; this holds for $N = \mathbb{R}^k \times \mathbb{H}^{m_1} \times \dots \times \mathbb{H}^{m_r}$, since geodesics of a Riemannian product are products of geodesics and each factor reflects across a geodesic). For $a = (t_a, \alpha)$ and $q = (t_0, \kappa)$ the minimising geodesic lies in the totally geodesic slab $\mathbb{R} \times \gamma$, where γ is the N -geodesic through α, κ ; consequently*

$$(t_a - t_0) \leq d(a, q) \leq (t_a - t_0) + \frac{d_N(\alpha, \kappa)^2}{2I(t_0, t_a)},$$

and $d(a, q)$ is a strictly increasing function of the intrinsic distance $d_N(\alpha, \kappa)$ at every scale and height. Hence the feet of a centre are exactly the d_N -nearest contacts, and a centre is medial iff its fibre position has two or more nearest contacts in the intrinsic metric of N .

Proof. The reflection r_γ of N across γ is an isometry fixing γ pointwise; $\text{id}_\mathbb{R} \times r_\gamma$ is then an isometry of $M_{f,N}$ whose fixed-point set is $\mathbb{R} \times \gamma$, hence totally geodesic and containing the minimiser. On $\mathbb{R} \times \gamma$ the metric is $dt^2 + f(t)^2 ds^2$ with s the N -arclength along γ — exactly the two-dimensional warped metric of Lemma 6.1 with horizontal gap $d_N(\alpha, \kappa)$ — so the bounds and strict monotonicity transfer verbatim. $\square$

Theorem 8.3 (Reconstruction over a Hadamard fibre). *Let N be geodesically reflective as in Lemma 8.2 (e.g. $N = \mathbb{R}^k \times \mathbb{H}^m$) and $K \subseteq S_0 = N$ closed.*

- (a) (Sharpness, unconditional.) *If K is convex in N , then $M_X = \emptyset$ and $\mathcal{R}_X = \emptyset$.*
- (b) (Sufficiency.) *If K is non-Chebyshev in N — some point has two or more nearest points — then $\{t > 0\} \subseteq \mathcal{R}_X$.*

Hence $\mathcal{R}_X \neq \emptyset \iff K$ is non-Chebyshev in N . If N has the Motzkin property (every closed non-convex set is non-Chebyshev), this reads $\mathcal{R}_X \neq \emptyset \iff K$ non-convex — the hyperbolic dichotomy, now on curved CAT(0) horospheres.

Proof. (a) A convex set in the Hadamard (hence CAT(0)) manifold N is Chebyshev: nearest-point projection onto a complete convex subset of a CAT(0) space is single-valued [5, II.2.4]. By Lemma 8.2 the foot of *any* centre $a = (t_a, \alpha)$ is the unique d_N -nearest contact to α (the monotonicity holds at every height), so $M_X = \emptyset$ and $\mathcal{R}_X = \emptyset$. (b) A point α^* with two d_N -nearest contacts k_1, k_2 gives, by Lemma 8.2, a two-footed centre (t_a, α^*) for every $t_a > 0$; the swallow estimate of Theorem 7.1 is purely radial (it uses only $I(t_p, t_a), I(0, t_a) \rightarrow \infty$) and transfers, giving $\{t > 0\} \subseteq \mathcal{R}_X$. $\square$

Two ingredients carry the sharpness proof: the foot-localisation of Lemma 8.2 (proved under geodesic reflectivity, and expected asymptotically for any Hadamard fibre), and the single-valuedness of projection onto a convex set — the only place the *curvature* of the horosphere enters, and precisely the CAT(0) property. So, granting localisation, the sharpness half of the dichotomy survives into curved horospheres exactly as far as CAT(0) reaches. The remaining content — sufficiency from non-convexity — is the Riemannian Chebyshev-set problem in N , Motzkin's theorem in $\mathbb{E}^{n-1}$ and open in full generality. The reconstruction theorem on $M_{f,N}$ is thus *equivalent* to the Motzkin property of the fibre N , isolating the entire difficulty in a single classical-type question about N alone. Necessity for general X is again the curvature-free deepest-shadow of Theorem 3.3 ($M_{f,N}$ is Hadamard), needing no separate shadow lemma; the sharpness statement (a) is unconditional.

9 Curved horospheres II: complex hyperbolic space

The first *positively* curved horosphere is the Heisenberg horosphere of $\mathbb{CH}^2$. Because the complex-hyperbolic distance is explicit, localisation — fact (L) of §3.1, elsewhere only expected — can here be carried out *exactly*, and it yields the first reconstruction theorem on a non-CAT(0) horosphere.

Horospherical coordinates. Realise $\mathbb{CH}^2$ as the negative lines in $(\mathbb{C}^3, \langle \cdot, \cdot \rangle)$ for the second Hermitian form $\langle z, w \rangle = z_1 \bar{w}_3 + z_2 \bar{w}_2 + z_3 \bar{w}_1$ of signature $(2, 1)$. A point with Heisenberg coordinates $(\zeta, v) \in \mathbb{C} \times \mathbb{R}$ and height $u > 0$ lifts to

$$\psi(\zeta, v, u) = \left(\frac{-|\zeta|^2 - u + iv}{2}, \zeta, 1 \right), \quad \langle \psi, \psi \rangle = -u.$$

The horospheres centred at the distinguished ideal point ($u \rightarrow \infty$) are the slices $\{u = c\}$, each a copy of the Heisenberg group $\mathfrak{H}_3 = \mathbb{C} \times \mathbb{R}$; the horoball is $\{u > c\}$.

Proposition 9.1 (Exact distance, and localisation to the Carnot metric). *For $p = (\zeta, v, u)$ and $p' = (\zeta', v', u')$,*

$$\cosh^2 \frac{d(p, p')}{2} = \frac{1}{uu'} \left[\left(\frac{|\zeta - \zeta'|^2 + u + u'}{2} \right)^2 + \left(\frac{v - v'}{2} + \operatorname{Im}(\zeta \bar{\zeta}') \right)^2 \right].$$

Writing $D = |\zeta_a - \zeta_q|^2$ and $V = v_a - v_q + 2 \operatorname{Im}(\zeta_a \bar{\zeta}_q)$, a high centre $a = (\zeta_a, v_a, u_a)$ and a contact q on $\{u = u_0\}$ obey the exact identity

$$\cosh^2 \frac{d(a, q)}{2} = \frac{u_a}{4u_0} + \frac{D + u_0}{2u_0} + \frac{(D + u_0)^2 + V^2}{4u_a u_0}.$$

Consequently the feet of a high centre are the contacts of least horizontal distance $|\zeta_a - \zeta_q|$, ties broken by least $|V|$: the vertical coordinate is suppressed by a factor $1/u_a$. The horizontal/vertical crossover is at $V \sim \sqrt{2u_a D}$, so under the parabolic dilation $(\zeta, v) \mapsto (\zeta, v/\sqrt{u_a})$ the nearest-point structure converges to that of the Carnot–Carathéodory metric on $\mathfrak{H}_3$ — not the Riemannian induced metric $4|d\zeta|^2/u_0 + \theta^2/u_0^2$, which would weight V at order one.

Proof. With $\psi_3 = \psi'_3 = 1$, $\langle \psi, \psi' \rangle = \psi_1 + \bar{\psi}'_1 + \zeta \bar{\zeta}' = -\frac{|\zeta - \zeta'|^2 + u + u'}{2} + i\left(\frac{v - v'}{2} + \operatorname{Im}(\zeta \bar{\zeta}')\right)$, and $\cosh^2(d/2) = \langle \psi, \psi' \rangle \langle \psi', \psi \rangle / (\langle \psi, \psi \rangle \langle \psi', \psi' \rangle) = |\langle \psi, \psi' \rangle|^2 / (uu')$, which is the first display. The second is its expansion in u_a at fixed u_0 (no approximation: $(D + u_a + u_0)^2 + V^2$ over $4u_a u_0$). The contact dependence is $\frac{D}{2u_0}$ at order 1 and $\frac{V^2}{4u_a u_0}$ at order u_a^{-1} ; equating them gives the crossover $V^2 = 2u_a D$. The induced horosphere metric follows from the infinitesimal form of the distance: $ds^2 = 4|d\zeta|^2/u_0 + \theta^2/u_0^2$ with $\theta = dv - 2 \operatorname{Im}(\zeta \bar{d}\zeta)$. (Numerically, as u_a runs $3 \rightarrow 1000$ the ambient-nearest of fixed contacts migrates from the Riemannian-nearest to the horizontal-nearest, and the crossover ratio $V^*/\sqrt{2u_a D} \rightarrow 1$.) $\square$

Theorem 9.2 (Reconstruction in $\mathbb{CH}^2$: sufficiency). *Let K be a closed contact set on a horosphere $\{u = u_0\}$ of $\mathbb{CH}^2$ that is graded non-Chebyshev: some position admits two contacts of equal least horizontal distance and equal V . Then the whole horoball interior $\{u > u_0\}$ lies in $\mathcal{R}_X$.*

Proof. Normalise $u_0 = 1$. Graded non-Chebyshev gives (ζ^*, v^*) and two contacts q_1, q_2 with equal $D = \delta^2$ and equal V , hence (Proposition 9.1) equidistant from $a_t = (\zeta^*, v^*, t)$ for every t , with no contact nearer; so a_t is medial for all large t . For an interior point $p = (\zeta_p, v_p, u_p)$, $u_p > 1$, the distance formula gives $\cosh^2(d(a_t, q_1)/2) = \frac{t}{4} + O(1)$ and $\cosh^2(d(p, a_t)/2) = \frac{t}{4u_p} + O(1)$, whence, via $\operatorname{arccosh}(x) = \log 2x + o(1)$,

$$r(a_t) = d(a_t, q_1) = \log t + O\left(\frac{1}{t}\right), \quad d(p, a_t) = \log t - \log u_p + o(1),$$

so $d(p, a_t) - r(a_t) \rightarrow -\log u_p < 0$. Thus $p \in B(a_t, r(a_t))$ with a_t medial for all large t , i.e. $p \in \mathcal{R}_X$. As p was arbitrary, $\{u > 1\} \subseteq \mathcal{R}_X$. $\square$

Proposition 9.1 settles localisation in $\mathbb{CH}^2$ exactly, with the intrinsic metric the Carnot one, and Theorem 9.2 is, to our knowledge, the first medial-axis reconstruction on a positively curved horosphere. The detection invariant is graded non-Chebyshev, read off the two feet, so it needs no Chebyshev-in-Carnot theory. What remains for the full $\mathbb{CH}^2$ characterization: necessity (a drifting medial centre forces graded non-Chebyshev); the exact reconstructed region for a general contact set; and, only for restating “graded non-Chebyshev” as a convexity, the right notion of convexity in $\mathfrak{H}_3$ (horizontal versus Euclidean).

10 The frontier: non-CAT(0) horospheres

The dividing line is now sharp. Over a flat fibre (M_f) the dictionary is literally Motzkin; over a nonpositively curved fibre ($M_{f,N}$, CAT(0) horospheres) the sharpness half is free and the dichotomy reduces to the Chebyshev property of the fibre. The first place where even the sharpness half can fail is a *positively* curved horosphere — $K_S > 0$ in some plane, which Proposition 4.1 permits under pinching and which no warped product over a Hadamard fibre can produce.

Rank-one symmetric spaces. In $\mathbb{CH}^n$ (and $\mathbb{HH}^n, \mathbb{OH}^2$, Damek–Ricci spaces) the isometry group still acts transitively on horoballs, so “WLOG one horoball” survives. But the horosphere is the Heisenberg group $\mathfrak{H}_{2n-1}$ with a left-invariant metric [7, 6, 9]: the geodesic up the axis contracts the $2n-2$ horizontal (contact) directions like $e^{-\tau}$ and the one vertical (centre) direction like $e^{-2\tau}$ — the anisotropy behind the $-4 \leq \sec \leq -1$ pinching of $\mathbb{CH}^n$. Its left-invariant metric has sectional curvatures of *both* signs (for $\mathfrak{H}_3$, Milnor’s $-\frac{3}{4}$ in the horizontal plane and $+\frac{1}{4}$ in planes through the centre [8]), so it is not CAT(0). For $n = 2$ both dictionary steps are carried out in §9 — localisation exactly, sufficiency from graded non-Chebyshev. In general n , localisation should give the same with the anisotropic two-rate Heisenberg metric (the analogue of Lemma 8.2 with a two-rate contraction in place of the single $1/I$), and detection becomes a Chebyshev-set problem in a Carnot group, sensitive to which convexity — Euclidean, geodesic, or horizontal — one means.

Pinched negative curvature. Dropping homogeneity, under $-b^2 \leq \sec \leq -a^2$ each horoball must be treated in place, its horosphere an Alexandrov space with two-sided curvature bounds (Proposition 4.1) and K_S generically of both signs. Localisation becomes a comparison statement — the discrimination weight lies between the Heintze–Im Hof rates $e^{-b\tau}$ and $e^{-a\tau}$, from Jacobi-field control of II — and detection is Motzkin’s theorem in an Alexandrov space of indefinite curvature, the precise admissible class being the open problem.

The stratification. The phenomenon is therefore controlled, at every level, by a single intrinsic question about the horosphere — whether a contact set with a two-footed centre is non-convex, and conversely — and the curvature of the horosphere governs how much of that equivalence survives:

- *flat* (M_f): Motzkin, both halves — full theorem (§7);
- CAT(0) ($M_{f,N}$): sharpness free, hard half = Motzkin in N — sharpness theorem (§8);
- *non*-CAT(0) ($\mathbb{CH}^2$): localisation and sufficiency via graded non-Chebyshev in the Carnot metric (§9);
- *non*-CAT(0), *general* ($\mathbb{CH}^n$, $n \geq 3$; pinched): two-rate / Alexandrov detection — open.

Flat and CAT(0) horospheres keep the sharpness half unconditionally; positive horosphere curvature is the genuine frontier.

Remark 10.1 (Prospects for the clean static form). The faithful characterization (Theorems 3.3–3.4) is already unconditional; what the frontier governs is only its *static* reading (Definition 3.5) — “defective = contact set non-Chebyshev in the intrinsic metric.” Localisation (L) is expected

throughout the pinched range by Jacobi-field comparison, with one subtlety: the intrinsic metric may be sub-Riemannian (Carnot) rather than Riemannian — a refinement of the *definition* of defective, not an obstruction, since the invariant is non-Chebyshev, which needs only a metric. The genuine nub is no-collapse (NC): on a positively curved horosphere the squared intrinsic distance loses 2-convexity past conjugate points, so two feet distinct at every finite height may coalesce in the limit. The cure, indicated by the hyperbolic experience, is to read defectiveness at *finite height*: two distinct feet at each t is all that medial means, and collapse in the limit is irrelevant, at the price of a per-height rather than closed-form criterion. Whether the closed-form static criterion itself survives in full positive-curvature generality reduces to one question: *can a limit-collapsing medial centre reconstruct?* The decisive test case is $\mathbb{CH}^2$, whose single homogeneous Heisenberg horosphere makes the computation explicit.

Conclusion. A characterization holds in full generality (Theorem 2.3): for every Hadamard manifold and every closed set, $\mathcal{R}_X$ is the hull together with the interiors of the defective maximal free horoballs, defectiveness read dynamically and the proof resting only on the curvature-free trichotomy and a curvature-free swallow lemma. Its faithful, M_X -free upgrade (Theorems 3.3–3.4) replaces the dynamical certificate by “the contact set differs from its horoball hull hconv,” which is conv on flat horospheres (Theorem 6 verbatim) and unconditional in every Hadamard manifold, necessity being Danskin’s deepest-shadow and sufficiency the trichotomy. What is special to constant curvature is not the theorem but the *static* reading of “defective”; the medial-axis reconstruction phenomenon is *not* special to constant curvature, with the curvature of the horosphere — not of the ambient manifold — governing how concrete that static reading can be made. On the warped family M_f the entire hyperbolic dictionary and the reconstruction dichotomy transfer verbatim despite variable pinched ambient curvature; warping over a curved Hadamard fibre gives CAT(0) horospheres on which the sharpness half remains an unconditional theorem; and the Heisenberg horosphere of $\mathbb{CH}^2$ gives the first reconstruction theorem on a positively curved horosphere. The genuine frontier is detection on non-CAT(0) horospheres in general: a Chebyshev question in a Carnot group, or Motzkin’s theorem in an Alexandrov space of indefinite curvature.

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